A simplified unified propagator model for signed order flow, concave price impact and rough volatility

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Abstract

In this note, we simplify the setup in [MORS26] by modelling persistent signed order flow F_t as the difference of two integrated hyper-rough Heston-type processes with an L^1-kernel and Hölder exponent in $(\frac{1}{2},1)$, which endogenously determines a (pure feedback) martingale price process P, the price impact kernel for the propagator equation, and the celebrated square-root law in the $H \to 0$ limit. This obviates the need to use nearly unstable heavy-tailed Hawkes processes, and lengthy macroscopic scaling limit arguments with C-tightness as in [MORS26]. We also discuss longer-memory/multiscale behaviour for F, and the addition of proportional bid-ask spreads and exogenous choice of price impact kernel, wherein P is no longer a martingale but the model is still arbitrage-free if P satisfies the conditional full support property in Guasoni et al.[GRS08]. We also propose alternate models where signed order flow is a Gaussian Volterra process with $H \in (\frac{1}{2},1)$, or using fGN which leads to power-law decay for the acf of order flow signs as in [LMF05],[MB25].

1.1 Persistent order flow

Following [AJ21] (and similar to Theorem 3.1 in [MORS26]), we model the cumulative positive order flow F_t^+ for an asset (i.e. the total number of shares bought by time t) as a weak solution to the Stochastic Volterra Equation:

$$F_t^+ \quad = \quad g(t) + \int_0^t f(t-s)M_s ds \tag{1}$$

where M is a continuous local martingale with $M_0 = 0$ and $[M]_t = F_t^+$ for $t \geq 0$, $f(t) = \nu e^{-\theta t} t^{H-\frac{1}{2}}$ with $H \in (-\frac{1}{4},0)$, $\theta > 0$ so $f \in L^1$, and $g(t) = V_0 t$ for some $V_0 > 0$ (note θ replaces the ergodicity parameter λ in Theorem 3.1 in [MORS26]). Weak existence and uniqueness for (1) is established in section 2 of [AJ21], and F^+ is a non-decreasing, nonnegative, continuous and adapted process. F_t^+ is also a.s. $(2\alpha - \varepsilon) \wedge 1$-Hölder continuous where $\alpha = H + \frac{1}{2}$ (see e.g. Theorem 3.1 in [JR20]), so from the condition on H we see that $2\alpha \in (\frac{1}{2},1)$ [1], and F_t^+ is non-differentiable if $H \leq 0$. Note that if $H > 0$, F^+ is just the integrated variance $\int_0^t V_s ds$ for a standard (i.e. non-hyper) Rough-Heston type variance process V, see e.g. section 5.4 of [FGS21] for details.

Lemma 1.1 $\mathbb{E}(F_t^+) < \infty$.

Proof. See Appendix A. ■

Corollary 1.2 *Assume (F^+, M) is a weak solution of (1) in the sense of [AJ21], with $b = 0$, $c = 1$ and $\nu \equiv 0$. Then M is a continuous square-integrable martingale.*

Proof. See Appendix B. ■

1.2 The inversion formula

If h is a function with $h * f \equiv 1$, then we have the inversion formula:

$$\begin{aligned}\int_0^t h(t-s)(F_s^+ - g(s))ds \quad &= \quad \int_0^t h(t-s)\int_0^s f(s-u)M_u du ds \quad = \quad \int_0^t \int_u^t h(t-s)f(s-u)ds\, M_u du \\ &= \quad \int_0^t \int_0^{t-u} h(t-s-u)f(s)ds\, M_u du \\ &= \quad \int_0^t (h*f)(t-u)M_u du \quad = \quad \int_0^t M_u du\end{aligned}$$

(where Fubini is justified since $F_t^+ < \infty$ a.s. for finite t and M is continuous), and hence

$$M_t \quad = \quad \frac{d}{dt}\int_0^t h(t-s)(F_s^+ - g(s))ds \quad = \quad \frac{d}{dt}\int_0^t h(t-s)F_s^+ ds - \frac{d}{dt}\int_0^t h(t-s)g(s)ds \tag{2}$$

[1] we can simulate F^+ using the Monte Carlo scheme in [AA25] using Normal Inverse Gaussian variates

Lebesgue a.e., where the final equality holds since $\int_0^t h(t-s)(F_s^+ - g(s))ds$ and $\int_0^t h(t-s)g(s)ds$ are both absolutely continuous, and hence so is their sum. Moreover, setting $\bar{H}(t) = \int_0^t h(s)ds$, integrating by parts we see that

$$\int_0^t H(t-s)dF_s^+ \quad = \quad F_s^+ \bar{H}(t-s)_{s=0}^{s=t} + \int_0^t h(t-s)F_s^+ ds \quad = \quad \int_0^t h(t-s)F_s^+ ds\,.$$

Taking Laplace transforms, the condition $h * f \equiv 1$ becomes $\hat{f}(\lambda)\hat{h}(\lambda) = \frac{1}{\lambda}$, from which we find that

$$h(t) \quad = \quad \frac{\theta^\alpha}{\nu\Gamma(\alpha)}(1 - \frac{\Gamma(-\alpha, t\theta)}{\Gamma(-\alpha)})$$

where $\Gamma(a,z) = \int_z^\infty s^{a-1}e^{-s}ds$ is the incomplete Gamma function, and $h(t) = O(t^{-H-\frac{1}{2}})$ as $t \to 0$. Moreover (from the Final value theorem for Laplace transforms), we see that

$$h(\infty) \quad = \quad \lim_{t\to\infty} h(t) \quad = \quad \lim_{\lambda\to 0} \lambda\hat{h}(\lambda) \quad = \quad \frac{1}{\hat{f}(0)} \quad = \quad \frac{1}{\|f\|_{L^1([0,\infty))}} \quad = \quad \frac{\theta^\alpha}{\nu\Gamma(\alpha)}\,. \tag{3}$$

1.3 Signed order flow

Again following [MORS26], we now assume the *signed* order flow for an asset is

$$F_t \quad = \quad F_t^+ - F_t^-$$

where $F_t^- = g(t) + \int_0^t f(t-s)M_s^- ds$ and (M^-, F^-) is an independent copy of (M, F^+). Then we see that

$$\frac{d}{dt}\int_0^t h(t-s)F_s ds \quad = \quad \frac{d}{dt}\int_0^t H(t-s)dF_s \quad = \quad \int_0^t h(t-s)dF_s \quad = \quad M_t - M_t^- \tag{4}$$

since the g terms cancel (and the second equality follows from the Stieltjes-Leibniz rule using the known Hölder continuity of $F^\pm$ and that $|h(u)| \le Cu^{-\alpha}$ so $\int_0^t |h(t-s)|(dF_s^+ - dF_s^-) < \infty$), and for $0 \le t \le u$ we have

$$\mathbb{E}(F_u^+|\mathcal{F}_t^M) \quad = \quad g(u) + \mathbb{E}(\int_0^u f(u-s)M_s ds\,|\mathcal{F}_t^M) \quad = \quad g(u) + \int_0^t f(u-s)M_s ds + M_t\int_t^u f(u-s)ds$$

(and similarly for $\mathbb{E}(F_u^-|\mathcal{F}_t^{M^-})$), since $(M_t)_{t\ge 0}$ is an $(\mathcal{F}_t^M)_{t\ge 0}$-martingale by Corollary 1.2, so

$$\mathbb{E}(F_u^+ - F_u^-|\mathcal{G}_t) \quad = \quad \int_0^t f(u-s)(M_s - M_s^-)ds + (M_t - M_t^-)\int_t^u f(u-s)ds \tag{5}$$

where $\mathcal{G}_t = \sigma(\mathcal{F}_t^M, \mathcal{F}_t^{M^-})$.

1.4 Endogenous price dynamics and the propagator equation

We now make the common assumption (cf.[JR20]) that the asset price $P_t = \kappa \lim_{u\to\infty} \mathbb{E}(F_u|\mathcal{G}_t)$ where $F_t = F_t^+ - F_t^-$ (for some $\kappa > 0$) [2]. Then letting $u \to \infty$ in (5), we find that $f(u-s) \searrow 0$ as $u \to \infty$ since $\theta > 0$ so the first term in in (5) goes to zero (from the monotone convergence theorem), but $\lim_{u\to\infty}\int_t^u f(u-s)ds = \lim_{u\to\infty}\int_0^{u-t} f(r)dr = \int_0^\infty f(r)dr = \|f\|_{L^1([0,\infty))} = 1/h(\infty)$ (from (3), so

$$P_t \quad = \quad \kappa \lim_{u\to\infty} \mathbb{E}(F_u^+ - F_u^-\,|\mathcal{G}_t) \quad = \quad \frac{\kappa}{h(\infty)}(M_t - M_t^-)$$

i.e. P is the difference of two i.i.d. driftless hyper-rough-type Heston models (which effectively both have correlation $\rho = 1$ since M and M^- only have one driving Brownian motion each so P is a "pure feedback" price process, i.e. only depends on its own history), and we recover the propagator equation:

$$P_t \quad = \quad \int_0^t G(t-s)dF_s$$

where the price impact kernel G is given by

$$G(t) \quad = \quad \kappa\frac{h(t)}{h(\infty)} \tag{6}$$

and we define the *unsigned* order flow as $V_t = F_t^+ + F_t^-$.

[2] note we are assuming $P_0 = 0$ without loss of generality to ease notation, but we can easily add a positive P_0 term

1.5 Power-law price impact

In particular, the *market impact function* of an exogenous metaorder executed at constant trading speed 1 up to time t_0 is given by

$$MI(t) \quad = \quad \int_0^{t\wedge t_0} G(t-s)ds \tag{7}$$

(where $\alpha_- = \frac{1}{2} - H$) which is $O(t^{\frac{1}{2}-H})$ as $t \to 0$ (for $H \neq \frac{1}{2}$) and globally concave in t for $H \in (-\frac{1}{2}, \frac{1}{2})$, consistent with empirical evidence (note the usual *square-root law* corresponds to $H = 0$ here), and (after cancelling terms) we can show that the asymptotic (i.e. permanent) impact as $t \to \infty$ (given no trading after t_0) is

$$MI(\infty) \quad = \quad \lim_{t\to\infty} \int_0^{t_0} G(t-s)ds \quad = \quad \int_0^{t_0} \lim_{t\to\infty} G(t)ds \quad = \quad \kappa t_0 \lim_{t\to\infty} \frac{h(t)}{h(\infty)} \quad = \quad \kappa t_0$$

(this corresponds to the asymptotic grey line in the plot below).

For the special case $H = \frac{1}{2}$ that we have previously excluded, $h(t) = \theta + \delta(t)$, which corresponds to linear permanent price impact plus temporary price impact; conversely $\lim_{\theta\to 0} h(t) = const. \times t^{-\frac{1}{2}-H}$. Recall from above that $(F_t^+)_{t\geq 0}$ has Hölder regularity $(2\alpha - \varepsilon) \wedge 1$, and empirical evidence in [MORS26] and below suggests that $2\alpha \in (\frac{1}{2}, 1)$ which corresponds to our standing assumption that $H \in (-\frac{1}{4}, 0)$.

1.6 Empirical results

Below we have tabulated estimates of 2α (i.e. the Hölder exponent of $(F_t)_{t\geq 0}$) using signed order flow for major US tech stocks using the model-independent $\hat{H}^{\pi}_{L,K}(X)$ estimator in Eq 7 in Cont&Das[CD24] which is consistent and allows for irregularly spaced data (with their $L/K = [\sqrt{N}]$, where N is the total sample size). For this we have used the free data at `https://data.lobsterdata.com/info/DataSamples.php`:

	GOOG	AAPL	MSFT	AMZN	INTC
N	49284	118326	410881	57360	404496
$2\hat{\alpha} = 2(\hat{H} + \frac{1}{2})$	0.692	0.720	0.847	0.735	0.890
$\hat{H}$	-0.154	-0.140	-.0767	-0.133	-0.0553

Note we only use columns 4 and 5 from the lobsterdata.csv files which correspond to actual realized trades.

Try Han-Schied SSE

1.7 Controlling liquidity

In the setup above, we have no control over the depth of liquidity, since the kernel $G(t)$ which controls the amount of transient price impact is endogenously determined via the f function in (1). To resolve this inflexibility, we can make the natural assumption that $P_t = \eta\tilde{M}_t + (1-\eta)\int_0^t G(t-s)dF_s$ for $\eta \in [0,1]$ where $\tilde{M}_t$ is an additional martingale, so η close to 1 implies greater liquidity and vice versa.

1.8 Using a mixed kernel

If we augment f to a function of the form $f(t) = e^{-\theta t}t^{H-\frac{1}{2}} + e^{-\theta_2 t}t^{H_2-\frac{1}{2}}$, then we find that

$$h(t) \quad = \quad \frac{1}{\Gamma(\alpha)} \sum_{n=0}^{\infty} (-c)^n \sum_{k=0}^{\infty} \frac{(\beta n)_k}{k!} \delta^k \theta^{-(\mu_n+k)} \frac{\gamma(\mu_n + k, \theta t)}{\Gamma(\mu_n + k)}$$

where $\beta = \frac{1}{2} + H_2$, $c = \Gamma(\beta)/\Gamma(\alpha)$, $\delta = \theta - \theta_2$, $\mu_n = \beta n - \alpha(n+1)$, $\gamma(a,z) = \int_0^z s^{a-1}e^{-s}ds$ is the analytic continuation of the lower incomplete Gamma function and $(a)_k$ denotes the Pochhammer symbol. We can then use two (possibly different) f functions to drive $F^{\pm}$, and if $H_2 \in (0, \frac{1}{2})$, P_t will have a conventional (i.e. non-hyper) rough component, which may be more realistic. We can also relax the assumption that M and M^- are independent to give greater control over the skewness of P_t.

1.9 Tempered long memory/multiscale behaviour

If we want $(F_t^{\pm})_{t\geq 0}$ to have longer memory/multiscale behaviour (as reported empirically in [MORS26]), we can replace f with e.g. $f(t) = (\frac{1}{c_1 t^{H-\frac{1}{2}}} + \frac{1}{c_2}t^{\gamma})^{-1}$ with $H \in (-\frac{1}{2}, 0)$, $\gamma > 1$ and $c_1, c_2 > 0$ (note f is still in L^1), in which case $P_t = \kappa\|f\|_{L^1([0,\infty))}(M_t - M_t^-)$.

1.10 Power-law decay for the acf of order flow signs

We can also consider a discrete-time formulation where signed order $F_n = \nu B_n^H$ for $n = 1, 2, ...$, where B^H is an fBM with $H \in (\frac{1}{2}, 1)$, and let $Z_n = B_n^H - B_{n-1}^H \sim N(0,1)$ so $(Z_n)_{n=1}^\infty$ is fractional Gaussian noise (fGN) with $Z_n \sim N(0,1)$ for all n, then

$$\begin{aligned} \rho(n) \;:=\; \mathbb{E}(Z_{k+n}Z_k) &= \mathbb{E}((B_{k+n}^H - B_{k+n-1}^H)(B_k^H - B_{k-1}^H)) \\ &= \tfrac{1}{2}((n+1)^{2H} - 2n^{2H} + (n-1)^{2H}) \;\sim\; const. \times n^{2H-2} \qquad (n \to \infty) \end{aligned}$$

and we see that Z is stationary. For two correlated $N(0,1)$ random variables X, Y with $\mathbb{E}(XY) = \rho$ it can be shown that $\mathbb{E}(\text{sgn}(X)\text{sgn}(Y)) = \frac{2}{\pi}\arcsin\rho$. Let $\varepsilon_n = \text{sgn}(Z_n)$ denote the sign of the change in order flow. Then

$$C(n) \;=\; \mathbb{E}(\varepsilon_k\varepsilon_{k+n}) \;=\; \frac{2}{\pi}\arcsin\mathbb{E}(Z_kZ_{k+n}) \;=\; \frac{2}{\pi}\arcsin\mathbb{E}(Z_0Z_n)\,.$$

But $\arcsin x = x + O(x^3)$ as $x \to 0$, so

$$\mathbb{E}(\varepsilon_k\varepsilon_{k+n}) \;\sim\; \frac{2}{\pi}\mathbb{E}(Z_kZ_{k+n}) \;\sim\; const. \times n^{2H-2}$$

as $|n| \to \infty$, so we have power-law decay for the acf of the change in order flow signs similar to the metaorder models in [LMF05] and [MB25].

1.11 Bid-ask spreads, exogenous choice of impact kernel, and non-semimartingale price dynamics

We can also include a proportional transaction cost $\varepsilon P_t > 0$ per share bought (i.e. a bid-ask spread). If we then also abandon the (questionable) assumption that $P_t = \kappa \lim_{u\to\infty}\mathbb{E}(F_u^+ - F_u^- \,|\mathcal{G}_t)$, and instead exogenously choose the price impact kernel G and assume that e.g. $P_t = P_0 + \int_0^{t\wedge\tau_\delta} G(t-s)dF_s$ for some deterministic function G where $\tau_\delta = \inf\{t : P_t = \delta\}$ and $P_0 > \delta > 0$ with $\delta \ll 1$; then if P has the conditional full support property, the model is arbitrage-free even if P is not a semimartingale (see e.g. Guasoni et al.[GRS08] for details). Note also that the trading costs due to G are quadratic.

1.12 Automated Market Makers

Let x and y denote the amounts two different coins in a liquidity pool. The basic Uniswap v1 protocol imposes that $xy = \kappa^2$ where κ is known as the *depth* of the pool, which is constant unless liquidity is added or removed from the pool by liquidity providers (LPs). If an Liquidity taker (LT) sells Δy coins to the pool, then the amount of asset X in the pool changes by amount $\frac{\kappa^2}{y} - \frac{\kappa^2}{y+\Delta y}$, and dividing this by Δy, and letting $\Delta y \to 0$, we see that the rate for an infinitesimal amount of Y is $z = \frac{\kappa^2}{y^2} = \frac{xy}{y^2} = \frac{x}{y}$. We can then assume that the amount of Y coins in the pool at time t is $Y_t = Y_0 + F_t^+ - F_t^-$ until Y hits zero, so $Z_t = \frac{\kappa^2}{Y_t^2}$. In practice, there are also proportional transaction costs of $\varepsilon = .003$, so as in subsection 1.11, if Z satisfies the conditional full support property, the model is arbitrage-free (note we are not assuming existence of an additional traditional LOB-based exchange like Binance here).

1.13 Signed order flow as a persistent Gaussian process

Let $\bar{F}(t) = \int_0^t f(s)ds$ where $f(.)$ is defined as in subsection 1.1 with $H \in (-\frac{1}{4}, 0)$, and we now assume the signed order flow F follows the Gaussian Volterra process:

$$F_t \;=\; \int_0^t \bar{F}(t-s)dW_s \;=\; \bar{F}(t-s)W_s|_{s=0}^{s=t} + \int_0^t f(t-s)W_s ds \;=\; \int_0^t f(t-s)W_s ds$$

where W is a standard Brownian motion. Then $\bar{F}(t) = O(t^{H+\frac{1}{2}}) = O(t^{H+1-\frac{1}{2}})$ as $t \to 0$, i.e. F has Hölder continuity $H + 1 - \varepsilon \in (\frac{1}{2}, 1)$, and

$$\mathbb{E}(F_u|\mathcal{F}_t^W) \;=\; \int_0^t f(u-s)W_s ds + W_t\int_t^u f(u-s)ds$$

and $\lim_{u\to\infty}\mathbb{E}(F_u|\mathcal{F}_t^W) = c_{H,\nu,\theta}W_t$ and $W_t = \int_0^t h(t-s)dF_s$ (interpreted as a Young integral), so the model is plausible but leads to boring Bachelier dynamics for the price process P_t (it does have the advantage that we can use MLE or GMM methods to estimate H and θ). Or we can replace W with another martingale (e.g. a process with rough volatility).

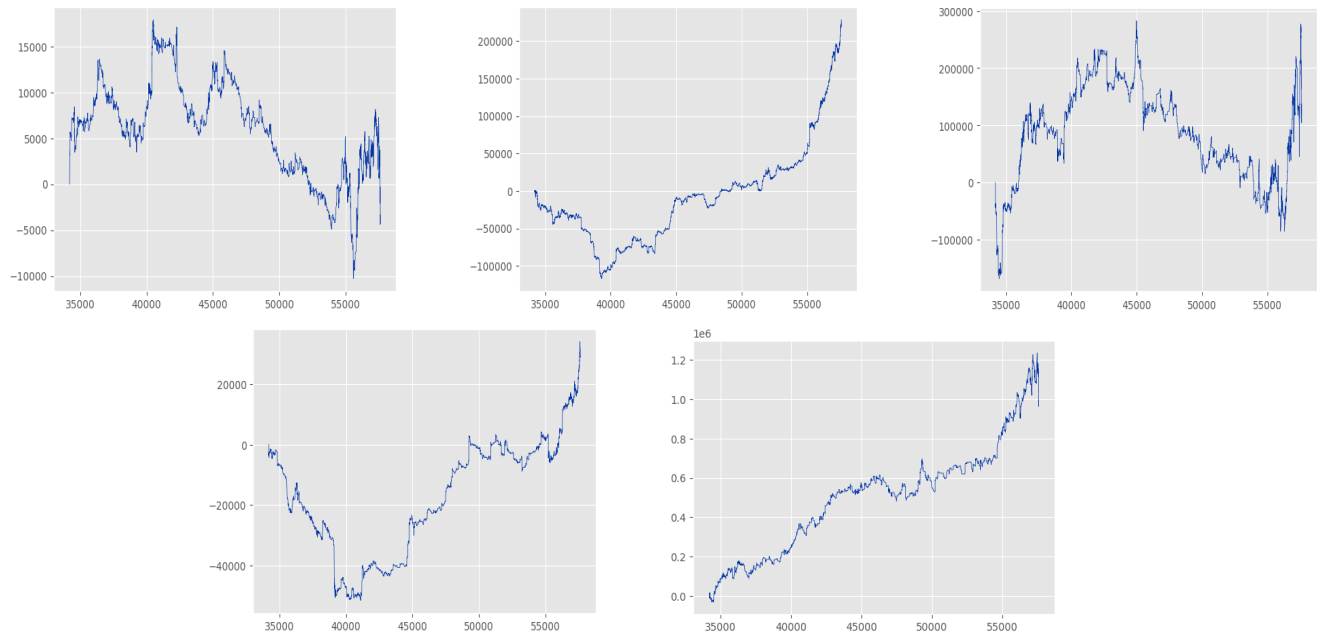

Figure 1: Signed order flow for GOOG, AAPL, MSFT, AMZN and INTC

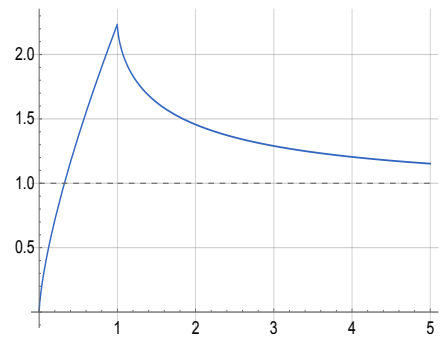

Figure 2: Concave price impact of an exogenous metaorder executed at constant trading speed 1 over $[0, t_0]$ with $H = -0.2$, $\theta = 0.1$, $\kappa = 1$ and $t_0 = 1$. The blue line asymptotes to the constant level $MI(\infty) = \kappa t_0 = 1$ in this case (grey line) as $t \to \infty$ which represents the asymptotic permanent price impact.

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A Appendix A

By Dambis–Dubins–Schwarz, there exists a Brownian motion B (on a possibly enlarged probability space) such that $M_t = B_{F_t^+}$. Now set $G_t := \sup_{0\le r\le t} |g(r)|$, $K_t := \int_0^t |f(r)|dr > 0$, and for $n \ge 1$, define the level hitting time $\tau_n := \inf\{s \ge 0 : F_s^+ \ge n\}$.

Since F^+ is continuous and nondecreasing, on $\{\tau_n \le t\}$ we have $F_{\tau_n}^+ = n$ and $F_r^+ \le n$ for all $0 \le r \le \tau_n$. Fix $n \ge 1$ and work on the event $\{\tau_n \le t\}$. For $0 \le r \le \tau_n$ we have $|B_{F_r^+}| \le \sup_{0\le u\le n} |B_u|$. Setting $s = \tau_n$ and taking absolute values we see that

$$\begin{aligned} n \;=\; g(\tau_n) + \int_0^{\tau_n} f(\tau_n - r)\, B_{F_r^+}\, dr \;&\le\; |g(\tau_n)| + \int_0^{\tau_n} |f(\tau_n - r)|\, |B_{F_r^+}|\, dr \\ &\le\; G_t + (\int_0^{\tau_n} |f(\tau_n - r)|\, dr) \sup_{0\le u\le n} |B_u| \\ &=\; G_t + (\int_0^{\tau_n} |f(s)|\, ds) \sup_{0\le u\le n} |B_u| \;\le\; G_t + K_t \sup_{0\le u\le n} |B_u|. \end{aligned}$$

Hence for $n > G_t$, on $\{\tau_n \le t\}$ we have $\{\sup_{0\le u\le n} |B_u| \ge \frac{n-G_t}{K_t}\}$. Since $F_t^+ \ge n$ iff $\tau_n \le t$, we obtain

$$\mathbb{P}(F_t^+ \ge n) \;\le\; \mathbb{P}(\sup_{0\le u\le n} |B_u| \ge \frac{n - G_t}{K_t}), \qquad n > G_t.$$

By the standard reflection principle bound $\mathbb{P}(\sup_{0\le u\le n} |B_u| \ge a) \le 4\exp(-\frac{a^2}{2n})$, for $a > 0$, so for $n > G_t$,

$$\mathbb{P}(F_t^+ \ge n) \;\le\; 4\exp(-\frac{(n - G_t)^2}{2nK_t^2}).$$

In particular, if $n \ge 4G_t$ then $(n - G_t)^2/n \ge \frac{1}{2}n$, hence

$$\mathbb{P}(F_t^+ \ge n) \;\le\; 4e^{-\frac{n}{4K_t^2}}, \qquad n \ge 4G_t. \tag{A-1}$$

Since $F_t^+ \ge 0$, we also know that $\mathbb{E}(F_t^+) = \int_0^\infty \mathbb{P}(F_t^+ \ge x)\, dx$. Now set $m := \lceil 4G_t \rceil$. Then

$$\mathbb{E}(F_t^+) \;\le\; m + \int_m^\infty \mathbb{P}(F_t^+ \ge x)\, dx \;=\; m + \sum_{n=m}^\infty \int_n^{n+1} \mathbb{P}(F_t^+ \ge x)\, dx \;\le\; m + \sum_{n=m}^\infty \mathbb{P}(F_t^+ \ge n). \tag{A-2}$$

Using (A-1) for all $n \ge m$,

$$\sum_{n=m}^\infty \mathbb{P}(F_t^+ \ge n) \;\le\; \sum_{n=m}^\infty 4\exp(-\frac{n}{4K_t^2}) \;<\; \infty. \tag{A-3}$$

Combining (A-2) and (A-3) yields $\mathbb{E}(F_t^+) < \infty$.

B Appendix B

By the weak-solution formulation, M is a continuous local martingale and $[M]_t = F_t^+$ for $t \ge 0$. Fix $t \ge 0$, and for $n \ge 1$ define

$$\tau_n := \inf\{s \ge 0 : [M]_s \ge n\}.$$

Then $(M_{s\wedge\tau_n})_{s\ge 0}$ is a continuous square-integrable martingale and (in particular) for every $0 \le s \le t$,

$$\mathbb{E}(M_{t\wedge\tau_n} \mid \mathcal{F}_s^M) = M_{s\wedge\tau_n}. \tag{B-1}$$

Moreover, by the Itô isometry for continuous martingales,

$$\mathbb{E}(M_{t\wedge\tau_n}^2) \;=\; \mathbb{E}([M]_{t\wedge\tau_n}) \;\le\; \mathbb{E}([M]_t) \;=\; \mathbb{E}(F_t^+) \;<\; \infty$$

where the last inequality follows from the previous lemma. Hence $(M_{t\wedge\tau_n})_{n\ge 1}$ is uniformly bounded in L^2, and therefore uniformly integrable. Since $[M]_t < \infty$ a.s. for fixed t, we have $\tau_n \to \infty$ a.s., and since M is continuous,

$$M_{t\wedge\tau_n} \to M_t, \qquad M_{s\wedge\tau_n} \to M_s$$

a.s. as $n \to \infty$. By uniform integrability, both convergences also hold in L^1. Passing to the limit in (B-1) yields

$$\mathbb{E}(M_t|\mathcal{F}_s^M) = M_s, \qquad 0 \le s \le t.$$

Thus M is a martingale. Finally, by Fatou's lemma,

$$\mathbb{E}(M_t^2) \;\le\; \liminf_{n\to\infty} \mathbb{E}(M_{t\wedge\tau_n}^2) \;\le\; \mathbb{E}(F_t^+) \;<\; \infty$$

so M is square-integrable.